

INA128 IC:

- Differential Pre-Amp Gain x10
- Gain = $1 + (50k/5.556k) = 10$.
- Reject common-mode noise (60 Hz line interfere)

High-Pass Filter:

- Cut-off frequency at 0.5 Hz to filter out the noise from muscle fiber under skin.
- $f_c = 1/(2\pi \cdot 680k \cdot 470n) = 0.5 \text{ Hz}$.

Low-Pass Filter:

- Cut-off frequency at around 40Hz to cut off the remaining 60Hz frequency from the INA128.
- $f_c = 1/(2\pi \cdot 18K \cdot 220n) = 40.2 \text{ Hz}$.

Final Gain Stage:

- Signal output is now around 10mV $\rightarrow$ 20 mV, so I need to increase the output signal for the MCU can read it.
- Gain = $1 + (100k/1k) = 101$
- Overall gain is $10 \times 101 = 1010$

AC Coupling & DC Bias:

- AC Coupling: Clean out the DC after the final gain stage
- DC Bias: Since the signal is fluctuating between the negative and positive value (which the MCU cannot read), I use DC Bias with +1.65V to crease the overall signal for it to fluctuate between 0V - 3.3V so that the MCU can record the signal without missing the negative values

